

ZIDIO INTERNSHIP PROJECT REPORT

**Project FORESIGHT Synthetic– Demand & Inventory
Intelligence**

**Demand Forecasting, Inventory Risk Scoring and Power BI
Planning Dashboard**

Data Science & Analytics | Zidio Internship

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Project Duration: 4 Weeks (1 Month)

2026

Executive Summary

Project FORESIGHT is a demand and inventory intelligence project designed around a retail business problem: balancing product availability with inventory efficiency. The Zidio project brief defines four central business asks—demand forecasting, stockout early warning, overstock identification, and a usable interface for non-technical decision-makers.

The project used four data extracts: daily sales, SKU master, calendar, and inventory snapshots. The analysis workflow covered data preparation, exploratory data analysis, demand forecasting, risk scoring, and business visualization. Multiple forecasting approaches were explored, including a baseline, Random Forest, XGBoost, LightGBM and Prophet, with the final prediction incorporated into the risk-scoring workflow.

The final analytical output combines forecast demand with inventory position to classify SKU-level inventory risk and recommend actions such as reorder, clearance/markdown, watch, or maintaining healthy inventory. A Power BI dashboard was developed to make the results interactive and accessible to business users.

Project Overview

Business Context

The client scenario describes a direct-to-consumer home and lifestyle business with approximately 200 active SKUs. Inventory planning was previously dependent on judgement and spreadsheets, creating two competing problems: stockouts of popular products and excess inventory of slow-moving products.

Problem Statement

The project aims to transform sales and inventory data into a SKU-level demand forecast and an early-warning decision system that identifies products requiring replenishment, products carrying excessive stock, and products that can be left unchanged.

Objectives

- Build an analysis-ready dataset from the four dataset extracts.
- Perform data-quality assessment and exploratory data analysis.

- Develop SKU-level demand forecasts and compare model predictions with a baseline.
- Convert forecast and inventory information into transparent stockout/overstock risk scores.
- Assign recommended actions for inventory planning.
- Develop an interactive Power BI dashboard.
- Present findings in a form understandable to non-technical decision-makers.

Project Requirements and Deliverables

The Zidio brief defines seven deliverables. The completed work in this report addresses the analytical and dashboard components developed during the project.

Deliverable	Requirement	Project Output
D1 Data pipeline	Ingest, clean and unify four extracts	Python/Pandas data preparation and merged analysis dataset
D2 Data quality & EDA	Identify data issues and business patterns	EDA, data-quality checks and business insights
D3 Demand forecast	SKU-level forecast with baseline comparison	Baseline + ML/forecasting models and final prediction
D4 Risk scoring	Risk level and recommended action per SKU	Inventory gap, risk score, risk category and action
D5 Planning dashboard	Interactive forecast/risk/reorder view	Power BI dashboard with KPIs, charts, slicers and action table
D6 Scoring service	Hosted forecast + risk service	Not evidenced in the supplied project materials
D7 Executive readout	Decision-focused stakeholder report	This project report

Data Description

- sales_daily: date, SKU, units sold, revenue, unit price and promotion flag.
- sku_master: SKU, category/subcategory, launch date, unit cost and list price.
- calendar: date, week, month, season, holiday indicator and promotion event.
- inventory_snapshots: date, SKU, on-hand units, on-order units, lead time and reorder point.

Data Preparation

The project involved checking structure, data types, missing values, duplicates and consistency across the four sources before combining the information for analysis. The final dataset was subsequently used for forecasting and inventory-risk analysis.

Data Quality

During the analysis, data-quality issues included missing values and duplicate/inconsistent records. The calendar data contained missing promotion-event values, which were treated as a data-quality issue rather than interpreted as a business event. Cleaning and validation were performed before the final dashboard dataset was produced.

Project Architecture

The project follows an end-to-end analytics workflow:

Raw Sales + SKU Master + Calendar + Inventory → Data Cleaning → EDA → Feature Engineering → Baseline → Forecasting Models → Final Prediction → Inventory Gap → Risk Score & Category → Recommended Action → Power BI Dashboard

Technology Stack

- Python
- Pandas and NumPy for data preparation and analysis
- Scikit-learn / machine-learning methods for modelling
- Prophet for time-series forecasting

- Jupyter Notebook for analytical development
- Power BI Desktop for interactive dashboarding

Execution Timeline

Week 1(20-07-2026 to 27-07-2026)

- Data Understanding and Data Preparation

Week 2(28-07-2026 to 03-07-2026)

- Exploratory Data Analysis
- Baseline and Feature preparation

Week 3(04-07-2026 to 11-07-2026)

- Forecast modeling and risk scoring

Week 4(12-07-2026 to 20-07-2026)

- Power bi Dashboard
- Executive Insights
- Finalisation and Recommendation
- Reporting

Exploratory Data Analysis

Key Dataset Metrics

- Total units sold: 868,789.
- Total revenue: approximately 151,721,667.33.
- Average units sold per record: approximately 11.90
- Average revenue per record: approximately 2,078.38.
- Sales and calendar data covered 2024; inventory snapshots covered a slightly shorter date range.

Sales and Category Analysis

The sales data was analysed by category, SKU and time to understand demand distribution and identify important product groups. The project included categories such as Furniture, Appliances, Decor and Kitchen.

Inventory Insights

- 1,082 inventory records were identified as stockout-risk records.
- These stockout-risk records involved 198 unique SKUs.
- SKU071 had the highest number of stockout-risk occurrences, with 12 occurrences.
- 6,799 inventory records were flagged as potential overstock using the screening rule $\text{On-hand Units} \geq 2 \times \text{Reorder Point}$.

Business Interpretation

The EDA indicates that inventory pressure exists in both directions. Repeated stockout-risk records suggest that some products require timely replenishment, while the potential-overstock screening identifies inventory that may tie up working capital. These findings support the need for a forecast-driven risk and action layer.

Demand Forecasting Methodology

Forecasting Approach

The project followed a multi-model forecasting workflow in which a baseline was established and additional models were explored.

Models Explored

- Baseline forecast.
- Random Forest regression.
- XGBoost regression.
- LightGBM regression.
- Prophet forecasting.

Feature and Forecast Workflow

The forecasting workflow used historical demand and engineered information to generate predictions. The final dataset contains `baseline_prediction`, `rf_prediction`, `xgb_prediction`, `lgb_prediction` and `final_prediction` fields. The final prediction was then used as an input to the inventory risk-scoring process.

Evaluation Principle

The forecasting models were evaluated using standard error metrics, including

- MAE (Mean Absolute Error)
- RMSE (Root Mean Squared Error)
- MAPE (Mean Absolute Percentage Error), to measure prediction accuracy.

In addition, WAPE (Weighted Absolute Percentage Error), MAE, and RMSE were calculated in Power BI using DAX measures and incorporated into the final dashboard for interactive evaluation of forecast performance. Lower values indicate better forecast accuracy.

Model Performance Comparison

Model	MAE	RMSE	MAPE
Baseline	4.7618715084	5.9770978178	52.2600721131%
Random Forest	0.0241459497	0.1285557125	0.2874891300%
XGBoost	0.2064674795	0.3426055932	1.8870386675%
LightGBM	0.2492410461	0.3977548734	2.2968807246%

Random Forest achieved the lowest MAE, RMSE and MAPE among the comparable SKU-level models and was therefore selected for the final demand prediction.

Prophet Evaluation

Prophet was evaluated separately using an aggregated daily demand series. Its MAE, RMSE and MAPE were recorded separately because its evaluation level differs from the SKU-level machine-learning models

Prophet was evaluated on aggregated daily demand and achieved MAE of 47.97, RMSE of 61.16, and MAPE of 2.03%; since it was not evaluated at SKU level, Random Forest was selected as the primary SKU-level forecasting model.

Forecasting Outcome

The modelling stage produced a final demand prediction for use in SKU-level inventory decisioning. The key practical contribution is the transition from

historical demand analysis to a forecast-informed inventory gap and risk assessment.

Inventory Risk Scoring

Risk Logic

The stockout risk is defined by comparing forecast demand over lead time with on-hand plus on-order stock, while overstock risk compares available stock with expected future demand. Each SKU should receive a risk level and recommended action.

Risk Variables

- SKU ID
- Category
- Final prediction
- On-hand units
- On-order units
- Inventory gap
- Risk score
- Risk category
- Recommended action

Inventory Gap

Inventory gap represents the difference between expected demand and available inventory position used by the project's risk logic. A negative or insufficient inventory position indicates potential replenishment pressure, while a large positive inventory position can indicate excess stock depending on the risk rule.

- If $\text{inventory_gap} < 0 \rightarrow \text{High Stockout Risk}$.
- If $\text{on_hand_units} \geq 2 \times \text{final_prediction} \rightarrow \text{Overstock Risk}$.
- Otherwise $\rightarrow \text{Healthy Inventory}$.

Risk score mapping: Healthy Inventory = 0; Overstock Risk = 1; High Stockout Risk = 2

Recommended Actions

Risk Category	Recommended Action
Overstock Risk	Reduce Inventory / Increase Promotion
Healthy Inventory	Maintain Current Stock
High Stockout Risk	Reorder Immediately

Business Value

Risk scoring converts a numerical forecast into an operational decision. The dashboard highlights which products require attention and what action is recommended.

Power BI Dashboard

Dashboard Purpose

The Power BI dashboard was designed as the stakeholder-facing layer of the project. It brings together demand predictions, inventory position, risk categories, risk scores and recommended actions so that users can filter and prioritise SKUs.

KPI and Visual Components

- Inventory/risk KPIs for monitoring the overall risk position.
- Risk-category donut chart to show the distribution of inventory conditions.
- Inventory position versus reorder-point analysis.
- Demand/forecast trend views.
- Category-level analysis.
- Recommended-action analysis.
- Interactive slicers for filtering the dashboard.
- SKU-level action table containing SKU ID, category, final prediction, risk category, risk score, inventory gap, recommended action, on-hand units and on-order units.

Action Table

The action table is one of the most decision-oriented elements of the dashboard. It provides the exact SKU-level information needed to move from analysis to action and supports prioritisation of products requiring replenishment or inventory reduction.

Dashboard Usability

The dashboard was designed to allow users to filter by relevant dimensions and inspect risk at both summary and SKU level.

Challenges and Solutions

Challenge	Solution / Handling
Missing or inconsistent data	Profiled fields, corrected data types and handled missing values during preprocessing.
Duplicate records	Checked for duplicates and cleaned records before downstream analysis.
Forecast model comparison	Used a baseline plus multiple modelling approaches before producing final predictions.
Inventory risk interpretation	Combined final prediction with inventory fields and converted results into risk/action fields.
Power BI data-field issue	Validated the final dataset and corrected the source/version issue so on-order and lead-time information became available.
Dashboard readability	Used slicers, KPIs, charts and an action table to support summary-to-detail analysis.

Recommendations

- Prioritize SKUs classified for immediate reorder and review their lead times before replenishment.

- Review overstocked SKUs for markdown, promotion or controlled clearance.
- Monitor repeatedly high-risk SKUs rather than treating each stockout event as an isolated incident.
- Use forecast and inventory gap together when making replenishment decisions.
- Refresh the dashboard and forecasting pipeline regularly as new sales and inventory data becomes available.
- Track forecast accuracy using WAPE against the approved baseline during future refreshes.
- Add a quantified rupee sales-at-risk and locked-capital measure if the required cost/price inputs are validated for the final dataset.

Alignment with Zidio Acceptance Criteria

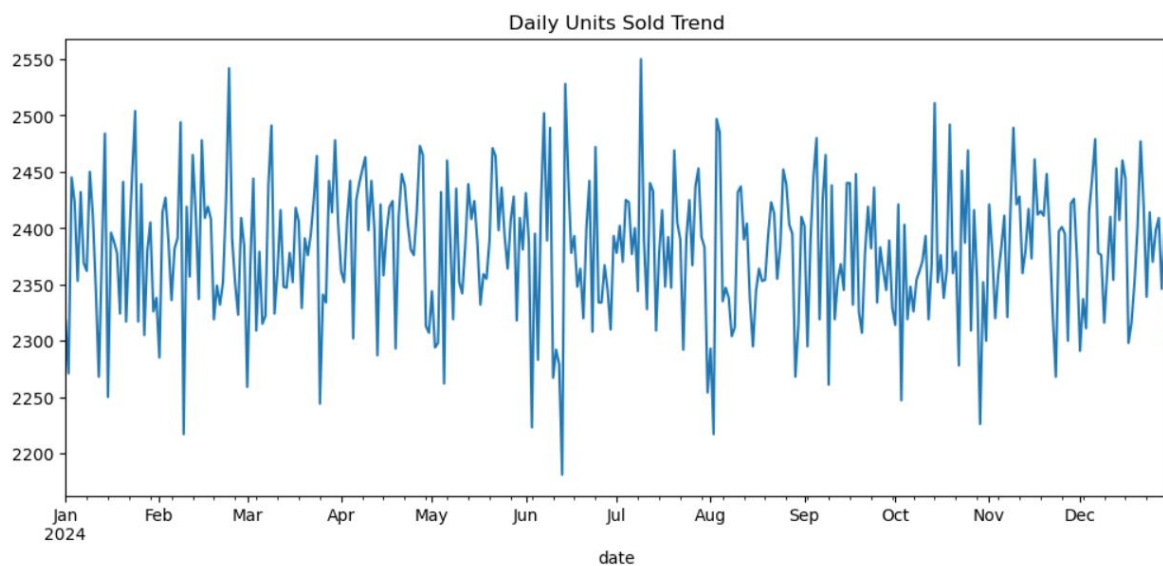
Requirement	Evidence in Project	Status
D1: Four extracts and analysis-ready data	Python/Pandas cleaning and merged final dataset	Completed
D2: Data quality and EDA	EDA metrics, stockout/overstock analysis and business insights	Completed
D3: Forecast	Baseline and multiple forecasting models; final prediction	Completed, but final WAPE evidence should be attached
D4: Risk scoring	Risk score, category, inventory gap and recommended action	Completed
D5: Planning dashboard	Power BI interactive dashboard with filters, visuals and action table	Completed
D6: Deployed scoring service	No verified URL/evidence in supplied materials	Not evidenced
D7: Executive readout	This report provides the project readout	Completed

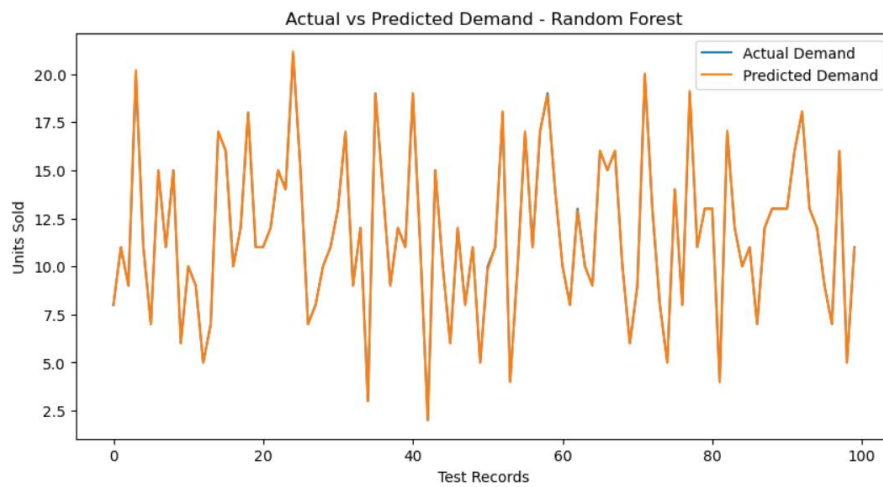
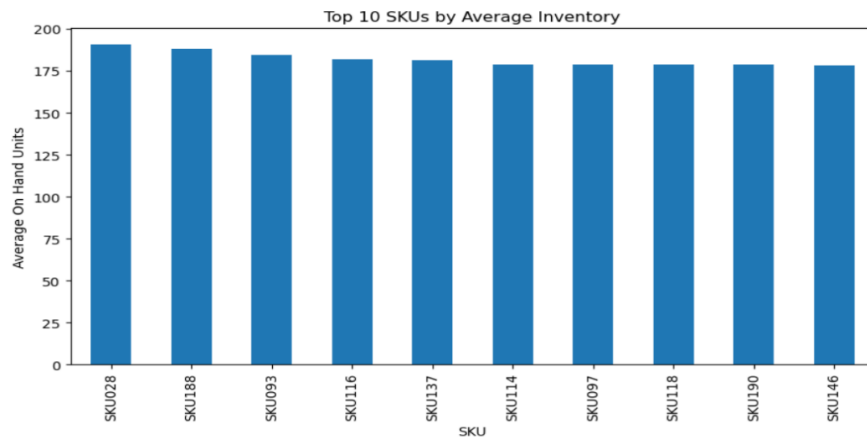
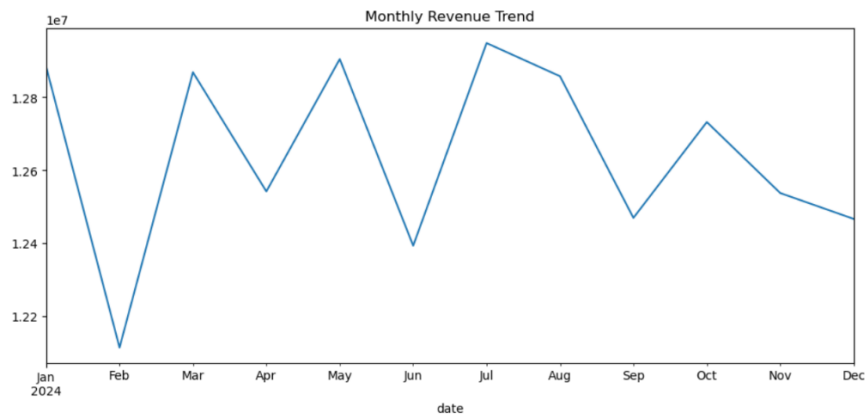
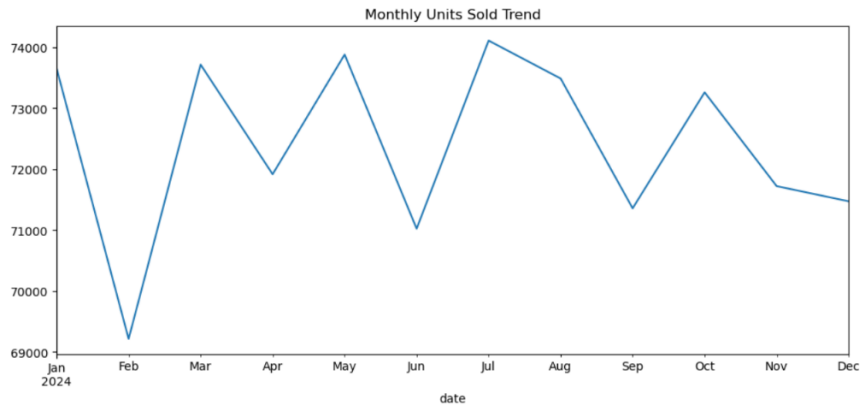
Personal Learning and Reflection

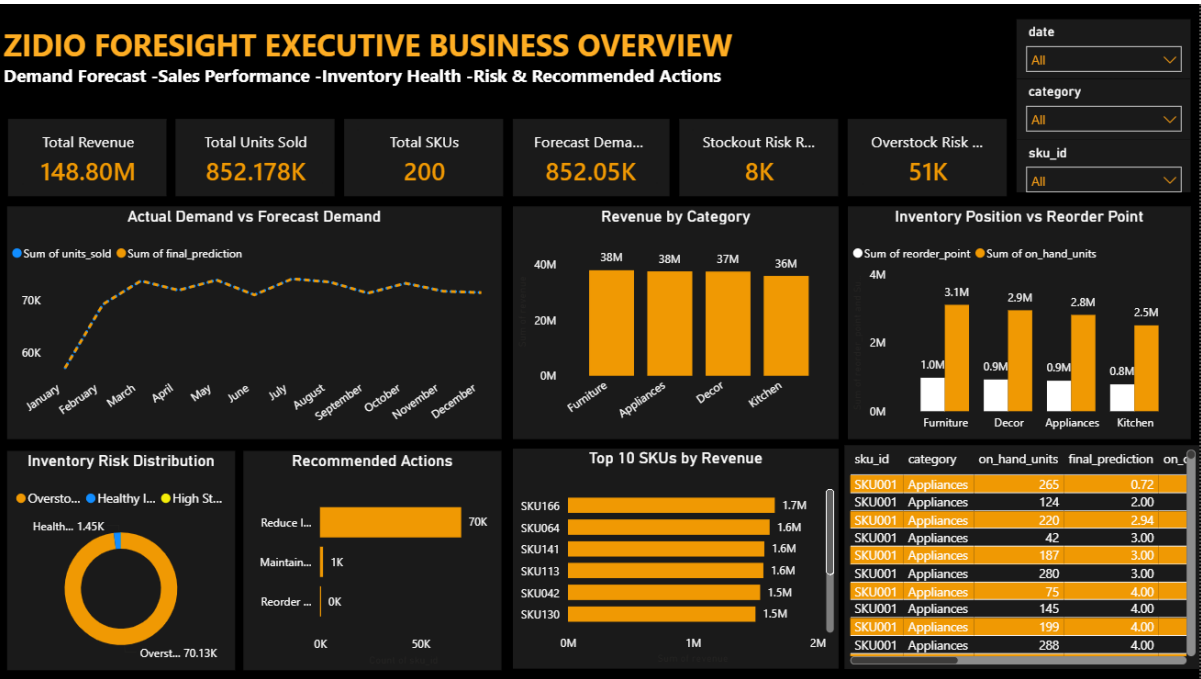
This project provided practical exposure to an end-to-end data analytics and data science workflow. The work progressed from raw retail extracts through data cleaning and exploratory analysis to forecasting, risk scoring and business dashboard development.

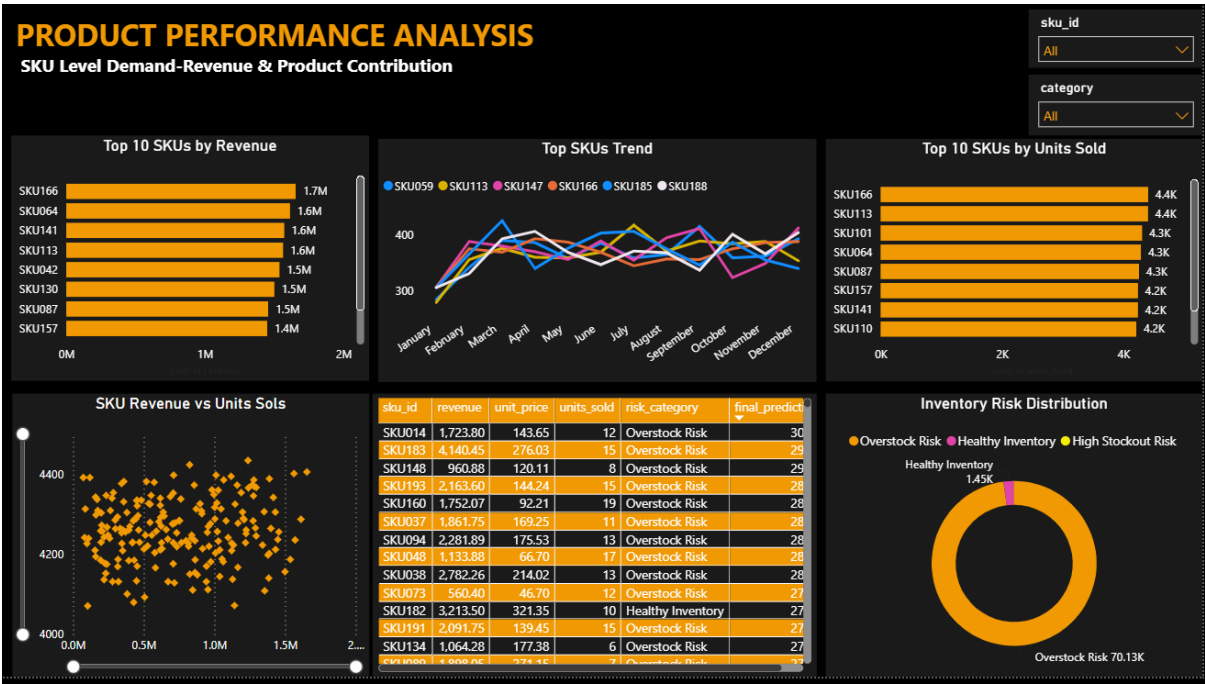
- Improved practical experience with Pandas-based data preparation.
- Learned how multiple retail datasets can be combined for business analysis.
- Gained experience with demand forecasting and model comparison.
- Learned how forecasts can be converted into transparent business rules.
- Developed practical Power BI dashboard and interactive-filtering skills.
- Improved ability to communicate technical analysis through business-oriented KPIs and recommended actions.

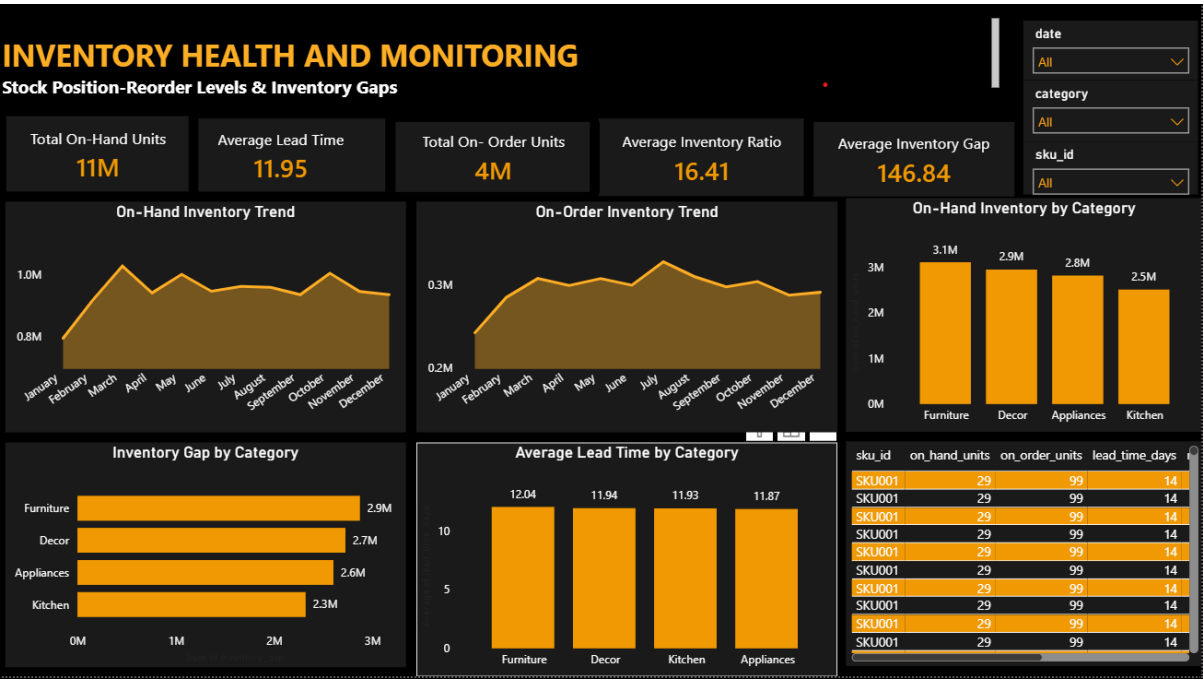
Visuals











Excess Inventory-Slow Moving Products & Clearance Priorities

A donut chart illustrating the distribution of inventory status. The chart is divided into three segments: a large orange segment for 'Overstock Risk' (70.13K), a small blue segment for 'Healthy Inventory' (1.45K), and a very small yellow segment for 'High Stockou...'. A legend on the right identifies the colors: orange for 'Overstock Risk', blue for 'Healthy Inven...', and yellow for 'High Stockou...'.

Category	Value
Overstock Risk	70.13K
Healthy Inventory	1.45K
High Stockou...	-

SKU	Sum of on_hand_units (orange)	Sum of reorder_point (blue)
SKU028	69K	19K
SKU188	66K	16K
SKU093	66K	18K
SKU146	66K	18K
SKU114	65K	18K
SKU008	65K	17K
SKU055	64K	16K

Category	Count
Furniture	20K
Decor	19K
Appliances	18K
Kitchen	16K

sku_id	on_hand_units	final_prediction	risk_score	reorder_point	reorder_point_min	reorder_point_max
SKU001	32	6.00	1	32	Reorder	Reorder
SKU001	32	10.00	1	32	Reorder	Reorder
SKU001	32	13.00	1	32	Reorder	Reorder
SKU001	32	14.00	1	32	Reorder	Reorder
SKU001	32	17.00	0	32	Max Stock	Max Stock
SKU001	37	8.00	1	75	Reorder	Reorder
SKU001	37	9.00	1	75	Reorder	Reorder
SKU001	37	11.00	1	75	Reorder	Reorder
SKU001	37	13.00	1	75	Reorder	Reorder
SKU001	37	14.00	1	75	Reorder	Reorder
SKU001	37	15.06	1	75	Reorder	Reorder
SKU001	42	3.00	1	60	Reorder	Reorder
SKU001	42	5.00	1	60	Reorder	Reorder

SKU	Count
SKU028	69K
SKU188	66K
SKU093	66K
SKU146	66K
SKU114	65K
SKU008	65K
SKU055	64K
SKU112	64K
SKU097	64K
SKU192	64K

Promotional Impact on Demand & Sales

148.80M

852K

11.90

71.6K

The chart displays the volume of tweets for the top 100 tweets (orange line) and the bottom 100 tweets (white line) over a 12-month period. The top 100 tweets maintain a high volume, fluctuating between approximately 10M and 15M tweets per month. In contrast, the bottom 100 tweets receive a much lower volume, consistently below 1M tweets per month.

Month	Top 100 Tweets (Approx. Count)	Bottom 100 Tweets (Approx. Count)
January	10M	0.5M
February	12M	0.5M
March	13M	0.5M
April	12M	0.5M
May	12M	0.5M
June	12M	0.5M
July	12M	0.5M
August	12M	0.5M
September	12M	0.5M
October	12M	0.5M
November	12M	0.5M
December	12M	0.5M

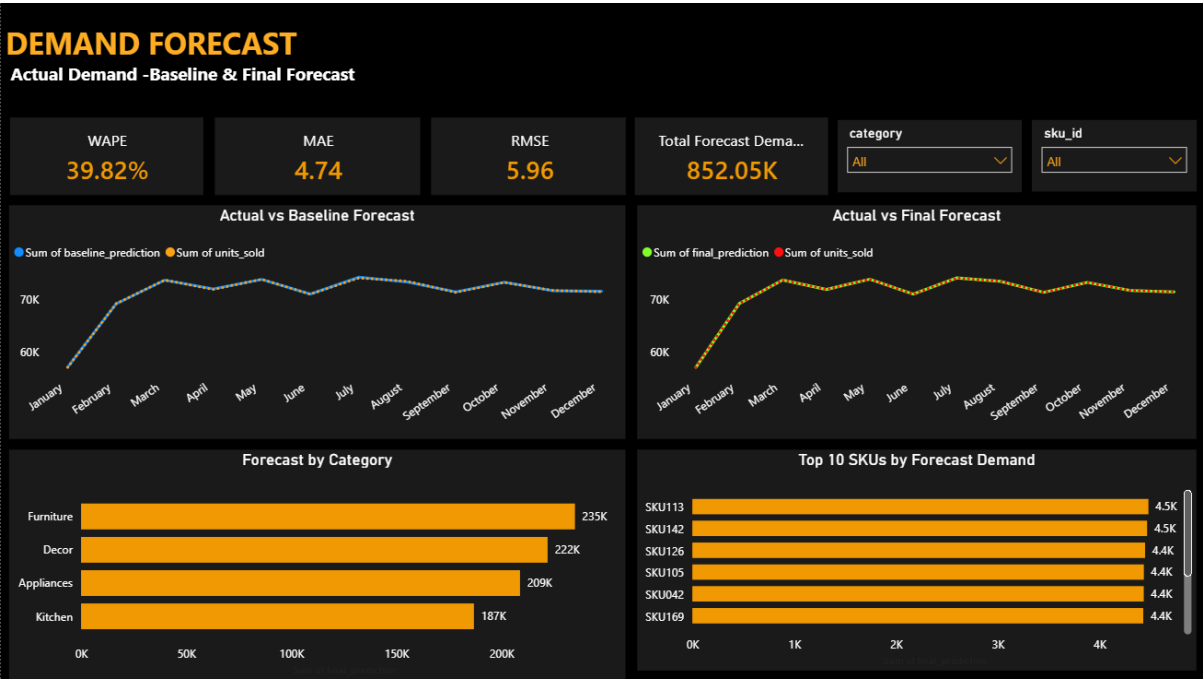
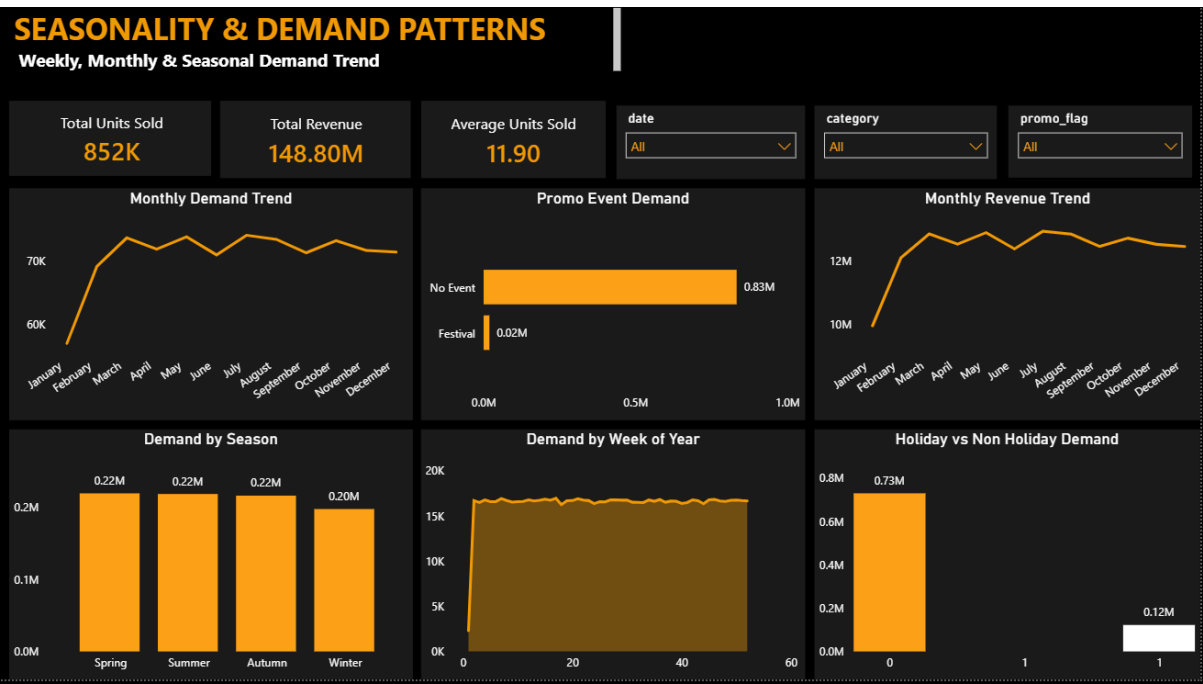
category	promo_flag	revenue	units_sold
Appliances	0	0.00	0
Decor	0	0.00	0
Furniture	0	0.00	0
Kitchen	0	0.00	0
Appliances	0	23.47	1
Appliances	0	39.55	1
Appliances	0	46.48	1
Appliances	0	52.21	1
Appliances	0	57.55	1
Appliances	0	66.27	1
Appliances	0	71.09	1
	0	71.46	

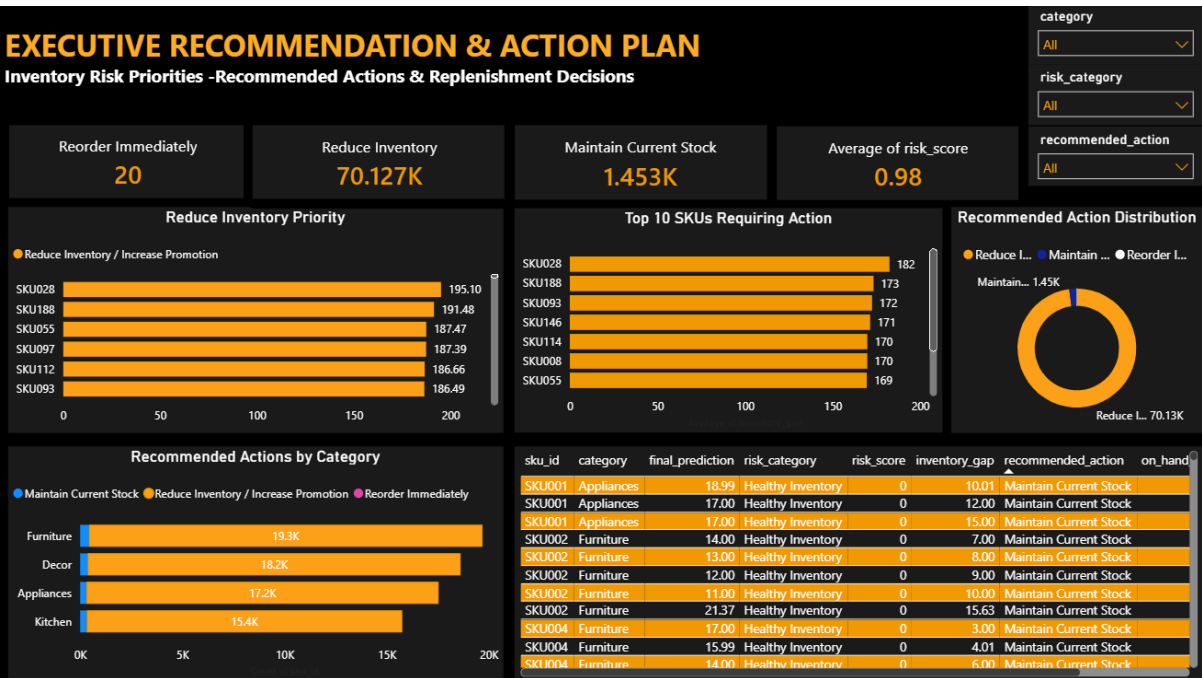
Category	Count
Furniture	0.21M
Decor	0.20M
Appliances	0.19M
Kitchen	0.17M

Category	Value
0	134M
1	15M

Category	Value
0	0.76M
1	0.09M

Age Group	Proportion (Yes)
18-24	0.00
25-34	0.00
35-44	0.03
45-54	0.11
55-64	0.35
65-74	0.74
75-84	0.89
85+	1.00





Conclusion

Project FORESIGHT demonstrates how retail sales and inventory data can be transformed into a practical demand and inventory intelligence solution. The project combines exploratory analysis, Feature engineering forecasting, inventory-gap analysis, risk scoring and Power BI visualization to support better stocking decisions.

The strongest business contribution is the connection between prediction and action: the final forecast is not presented in isolation, but is combined with inventory information to identify risk and recommend what should be reviewed. The resulting Power BI dashboard provides an accessible interface for monitoring the inventory position and prioritizing SKU-level actions.